METEOR: Evidence-weighted reaction selection reconciles enzyme prediction with growth feasibility (Supplementary material)

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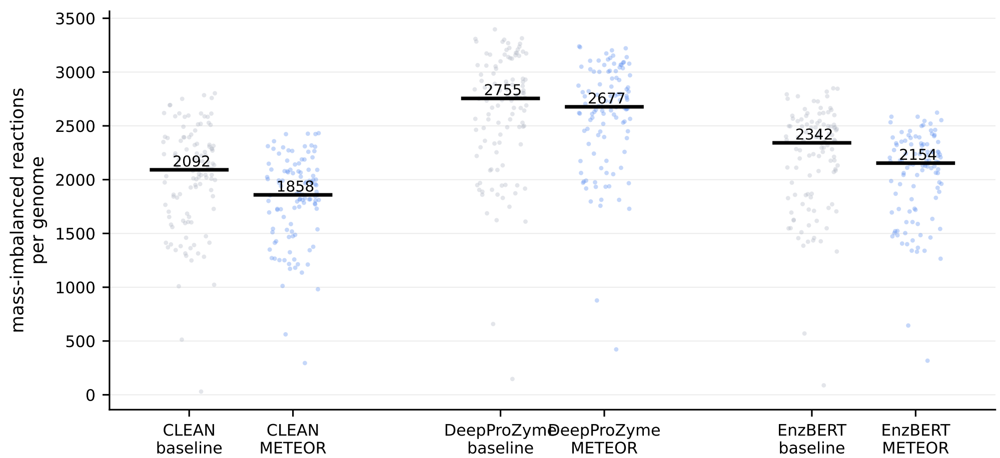

Fig. S1. Mass-imbalanced reactions per genome across the 108-GCF panel, measured identically on both sides with `find_mass_unbalanced_reactions`. Baseline threshold networks (τ=0.5) have medians of 2,092 (CLEAN), 2,755 (DeepProZyme) and 2,342 (EnzBERT); the corresponding METEOR networks have medians of 1,858, 2,677 and 2,154. METEOR is lower for every predictor, but the submodels also differ in size (DeepProZyme: 5,431→5,671 reactions), so the mass-imbalanced fraction in main text Table 1 is the size-normalised comparison. Each point is one genome; horizontal bars mark medians. Generated by `eval/gen_fig_massimbal.py` from the same per-genome records as main text Table 1.

S1. Parameters

Figure S1 shows the per-genome mass-imbalance distributions behind main-text Table 1. Default values used throughout the study are: $p = 2$ and $\mu = 3$ (exponent and weight of the evidence-weighted parsimony penalty $\mu(1-w_j)^p$ added to the log-odds cost; the distinguishing term in Eq. (2) of the main text); $\gamma_{\min} = 0.1$ (lower bound on the selected Gram-specific SEED biomass reaction); $\lambda = 10^{-4}$ (parsimony weight on flux $|v|$); $\alpha = 0.4$ (per-EC confidence-update step size in the update equation of main text, Section 2.2; the code exposes this as `--beta`, with $\alpha = \min(1, 0.4\beta)$, so the default β=1 is the α=0.4 quoted here); $\epsilon = 10^{-6}$ (log-odds smoothing in the cost equation); $\delta = 0$ (forced-flux floor, disabled; a distinct parameter that would require $|v_j| \geq \delta$ for every selected reaction); decision threshold $\tau = 0.5$ when a hard EC call is required downstream (e.g. for pathway completeness).

Reactions with no EC annotation receive no special treatment. The implementation clips their confidence from zero to $w_j = 10^{-6}$ before applying the cost equation, giving a cost of approximately 16.1 at these defaults. Apart from fixed medium reactions, this strongly disfavours their selection. At $w_j = 0.5$, the cost is 0.75; it crosses zero at approximately $w_j = 0.611$.

We did not tune $\gamma_{\min}$, λ or α to maximise any metric on the holdout: all three were fixed during earlier development on E. coli and frozen before any holdout evaluation. Sensitivity to these choices is reported in Section S5 of this document.

We classify organisms as Gram-negative or Gram-positive by NCBI taxonomy, which selects both the SEED biomass reaction variant and a set of precomputed tight flux bounds obtained by flux-variability analysis on the full universal model under the defined medium.

S1.1 Complete optimisation protocol

The following steps run in `eval/emit_v8.py` between the score matrix and the MILP. We list them so the pipeline can be reproduced end to end.

Input guard. We skip a genome when its baseline prediction matrix has fewer than $0.9\times$ the row count of the EnzBERT matrix for the same assembly. This is the exclusion criterion behind the 108-genome panel.

Score truncation. Each protein keeps its five highest-scoring ECs; all other entries become zero before the noisy-OR of main text Eq. (1). Retaining the full row lets the many small nonzero scores that some predictors assign per protein saturate w_j.

Stoichiometry and bounds. We build $\mathbf{S}$ from the SEED universal with reversible transport reactions expanded, then intersect the universal bounds with the Gram-specific tight bounds above: $lb \leftarrow \max(lb, lb_{\text{tight}})$ and $ub \leftarrow \min(ub, ub_{\text{tight}})$.

Excluded reactions. Reactions that cannot carry flux under those bounds cannot be selected. We also exclude the biomass reaction for the other Gram class; the biomass reaction used as the objective remains available.

Biomass-feasible skeleton. We solve an LP maximising v_{biomass} subject to $\mathbf{Sv} = \mathbf{0}$ and the bounds, then take every reaction with $|v_j| > 10^{-6}$ in that solution as the skeleton. The LP never sees the genome's scores, so the skeleton is a property of the Gram group and medium rather than of the organism.

Candidate mask. Reaction j is selectable if $w_j \geq w_{\min} = 0.01$, or j lies in the skeleton, or j is a medium exchange; excluded reactions are then removed. Every remaining y_j is fixed to zero, which is what keeps the MILP tractable on a 47,880-reaction universal.

MILP bounds and stopping rules. $\gamma_{\min} = 0.1$ (biomass floor); $\gamma_{\max} = 2.5$ (upper bound on normalised biomass flux); $\lambda = 10^{-4}$; and forced-flux floor $\delta = 0$ (disabled). CBC runs with four threads, a 600 s limit and `gapRel`$= 0.05$. We record the returned `LpStatus` and the wall time around the solve call; on the DeepProZyme-vanilla panel this gives Optimal 107/108 and Not Solved 1/108.

The MILP operates under a defined medium: 36 exchange reactions representing essential nutrients (the `minimal_uptake` set in the codebase) have their uptake bounds set to the universal model's defaults, while all other exchange reactions are blocked for uptake (lb=0) but may carry secretion flux. The post-solve growth check uses this same medium. MEMOTE and the additional downstream FBA check use a separately inferred uptake medium, as described in Section S7.1.

The verify-and-repair step tests whether reactions with $y_j > 0.5$ support $\max v_{\text{biomass}} \geq 0.05$ in a standalone FBA. Approximately 7.5% of raw solutions across the evaluated configurations fail this check because near-zero binary values can carry flux through large bounds within the solver's numerical tolerance. For these solutions, an LP adds the support that minimises flux through reactions not already selected while satisfying the biomass floor. Every model reported here passed the post-repair check.

Solver protocol and solution status. The reported MILPs were solved with CBC (`PULP_CBC_CMD`) using four threads, a relative optimality gap `gapRel`$= 0.05$ (5%), and a 600 s time limit. We set no random seed, and four-threaded CBC does not guarantee bit-identical re-runs: degenerate optima of equal objective value can differ by a small number of reactions between runs. The released implementation can also select Gurobi or HiGHS when available, but the reported runs used CBC. A feasible incumbent satisfies the numerical formulation: mass balance $\mathbf{Sv} = \mathbf{0}$, inclusion-gated bounds $L_j y_j \leq v_j \leq U_j y_j$, and $v_{\text{biomass}} \geq \gamma_{\min}$. Unless a solve reaches the 600 s limit, the returned solution is within the 5% relative gap of the MILP optimum. All reported selected sets were independently checked for growth after repair.

The big-M formulation admits degenerate optima in which the selected set does not itself carry biomass flux, because the inclusion gate $v_j \leq U_j y_j$ leaves a reaction able to carry negligible flux at y_j below the solver's integrality tolerance. We therefore re-maximise biomass over each selected set alone, under the medium and bounds the MILP used, and repair from the same candidate set where it falls short; three genomes in the deposited DeepProZyme-vanilla solution set carry such a repair, adding 59, 64, and 88 reactions. Every deposited selected set clears the floor on its own after repair, the lowest at 0.38 against $\gamma_{\min}$=0.1. The deposited `y_vals` therefore represent the repaired set. The deposited `v_vals` and `biomass_flux` retain the original MILP witness and should not be read as a flux witness for reactions added during repair. Structural results were recomputed from the repaired submodels.

S2. Retraining

To ensure systematic evaluation, we retrain baselines at four sequence-identity cutoffs: 0.30, 0.50, 0.70, and an unfiltered baseline (denoted vanilla, i.e., the original publicly released checkpoint). At a specified cutoff X, we eliminate all proteins that share $\geq X$% identity with the 227 holdout proteins from the training pool. We retrain CLEAN on its own filtered split100 data. EnzBERT and DeepProZyme are retrained on SwissProt 2021_04 to ensure parity between those two baselines. All three baselines exclude holdout proteins at each sequence-identity cutoff. Repository URLs for the training splits and code are provided in the Reproducibility section of the main text.

The monotone drop in baseline accuracy from vanilla to filt70 to filt50 to filt30 (e.g. EnzBERT top-1 41 $\rightarrow$ 38 $\rightarrow$ 35 $\rightarrow$ 22 on the full n=125 SwissProt holdout, before genome-resolution into the 108-GCF panel; on the smaller n=66 genome-resolved subset only EnzBERT falls steadily across the cutoffs (24 → 23 → 21 → 17), DeepProZyme is flat at 17, and CLEAN varies without a clear direction, which the smaller sample makes noisy; Table S1) is expected: stricter filtering removes more training examples with close similarity to the holdout proteins, leaving the model with fewer examples of the exact function families that appear in the holdout. The large step at the 30% cutoff (filt50 35 → filt30 22 for EnzBERT) is consistent with the 30%–50% band containing a disproportionate share of the nearest-neighbour training exemplars for holdout proteins.

S2.1 Filtered-retraining holdout-66 results

Table S1 reports protein-level fidelity on holdout-66 across all four sequence-identity retraining cutoffs. filt30/filt50/filt70 denote baselines retrained after removing training proteins sharing $\geq 30/50/70\%$ identity with the 227 holdout proteins. Under the net-only comparison (baseline and METEOR mapped through the identical EC-vocabulary normalisation), top-1 is exactly preserved in all 12 configurations. METEOR changes no top-1 call anywhere, so the paired test has no discordant pair to work with and we report none, while $F_{\max}$ stays within ± 0.012 with no consistent sign. In this holdout, the update therefore changes no top-1 call across predictors and retraining cutoffs. The small positive top-1/$F_{\max}$ gains reported by a naive (non-net-only) comparison are an EC-vocabulary artifact (Section S8).

Table S1. Protein-level fidelity on holdout-66 across all baseline×retraining-cutoff configurations, net-only (baseline and METEOR mapped through the identical EC-vocabulary normalisation). B = baseline; $+M$ = METEOR-refined.

	top-1 ↑		$F_{\max}$ ↑		
Baseline	B	$+M$	B	$+M$	$\Delta F_{\max}$
CLEAN-vanilla	26	26	0.429	0.428	±0.000
CLEAN-filt30	22	22	0.364	0.364	±0.000
CLEAN-filt50	25	25	0.421	0.421	±0.000
CLEAN-filt70	28	28	0.453	0.453	±0.000
DeepProZyme-vanilla	17	17	0.308	0.307	−0.001
DeepProZyme-filt30	17	17	0.318	0.317	±0.000
DeepProZyme-filt50	17	17	0.318	0.319	±0.000
DeepProZyme-filt70	17	17	0.314	0.314	±0.000
EnzBERT-vanilla	24	24	0.385	0.385	±0.000
EnzBERT-filt30	17	17	0.292	0.288	−0.004
EnzBERT-filt50	21	21	0.340	0.340	+0.001
EnzBERT-filt70	23	23	0.376	0.365	−0.012

Note. n=66 proteins. Top-1 is identical in every configuration, so no protein changes status and there is nothing for a paired test to compare. $\Delta F_{\max}$ lies within $[-0.012, +0.001]$, without a consistent sign.

S2.2 Calibration analysis

Table S2 reports calibration metrics on the same holdout-66 set, under the evidence-weighted cost used throughout the paper. We report it to document that the monotone per-EC update does not target calibration, and that where it changes calibration it does so unevenly across predictors.

Table S2. Calibration on holdout-66 (n=66). B = baseline; $+M$ = METEOR-refined. Bold indicates improvement. ECE (15 bins) improves in 6 of 12 configurations and Brier in 8. Deposited as `results/holdout66_calibration.json`.

	ECE ↓		Brier ↓	
Configuration	B	$+M$	B	$+M$
CLEAN-vanilla	0.140	0.151	0.129	0.136
CLEAN-filt30	0.301	0.299	0.279	0.278
CLEAN-filt50	0.319	0.309	0.275	0.274
CLEAN-filt70	0.333	0.334	0.307	0.309
DPZ-vanilla	0.436	0.427	0.358	0.344
DPZ-filt30	0.326	0.271	0.274	0.248
DPZ-filt50	0.332	0.290	0.289	0.266
DPZ-filt70	0.340	0.296	0.286	0.261
EnzBERT-vanilla	0.283	0.295	0.257	0.253
EnzBERT-filt30	0.130	0.158	0.143	0.154
EnzBERT-filt50	0.199	0.200	0.199	0.196
EnzBERT-filt70	0.305	0.339	0.291	0.299

Calibration changes are predictor-specific rather than systematic. ECE improves in 6 of 12 configurations and Brier in 8. DeepProZyme, whose initial miscalibration is the highest (ECE $\approx$0.3–0.4), improves on both metrics at all four cutoffs. CLEAN improves on both at filt30 and filt50 and worsens on both at vanilla and filt70. EnzBERT worsens on ECE at all four cutoffs while its Brier score improves at two. The confidence update is monotone within each EC column and does not target calibration, so it can move the two metrics in opposite directions, as it does for EnzBERT. We make no calibration claim.

S3. Ablation and controls

We evaluate METEOR's components by ablation on the 108-GCF panel, DeepProZyme-vanilla, using the fair net-only protein comparison and the pathway-detection metric of main text, Section 3.3. All conditions use the hard-biomass MILP unless stated otherwise.

Conditions. B0: baseline only (no refinement). B2: full MILP with the biomass floor removed ($\gamma_{\min}$=0). B3: full METEOR (evidence-weighted penalty, p=2, μ=3). uniform (p=0): the same hard-biomass MILP with a confidence-independent parsimony penalty ($\mu(1-w_j)^0$), isolating the contribution of evidence weighting. B2 and B3 share the same noisy-OR confidence w_j and log-odds cost; B0 supplies the same baseline scores but does not solve the MILP. B2 and B3 differ only in the biomass floor; B3 and uniform differ only in the penalty exponent p. The strongest control on the sequence-derived cost is the internal deletion-recovery experiment under evidence-derived, uniform, and shuffled costs (main text, Section 3.4), which quantifies off-reference selection while holding the feasible candidate set fixed.

Table S3. Protein-level ablation on the holdout-66 evaluation set (n=66, DeepProZyme-vanilla, net-only).

Condition	top-1 (/66) ↑	$F_{\max}$
B0 (baseline only)	17	0.308
B2 (no biomass floor)	17	0.307
B3 (full METEOR)	17	0.307
uniform (p=0)	16	0.305

Note. Net-only. B2/B3 preserve baseline top-1;
the confidence-independent uniform penalty drops one call.

Table S4. System-level ablation: pathway detection (P=151 SEED-representable pathways) on the 108-GCF panel, DeepProZyme-vanilla. Detected = mean pathways/genome with ≥1 EC / ≥50% of ECs present ($B \to M$). sub-thr. EC: fraction of available nonzero sub-threshold ECs (0 < score < 0.5) that the condition selects.

Condition	≥1 EC	≥50%	sub-thr. EC
B0 (baseline only)	123.9 (–)	16.6 (–)	–
B2 (no biomass floor)	123.9→132.2	16.6→35.3	18.6%
B3 (full METEOR)	123.9→132.5	16.6→36.0	19.1%
uniform (p=0)	123.9→121.9	16.6→22.6	10.8%

Note. evw (B3) detects +8.6 pathways at ≥1 EC; the
confidence-independent uniform penalty loses 2, pruning more nonzero signal.

Ablation findings. At the protein level (Table S3, net-only), B0, B2, and B3 have identical top-1 (17/66), while $F_{\max}$ changes from 0.308 to 0.307. The uniform penalty is the exception: it drops one top-1 call (16/66).

At the system level (Tables S4 and S5), the two penalties produce different selected sets. evw (B3) detects +8.6 pathways per genome at ≥1 EC and roughly doubles the ≥50% count, whereas the uniform penalty loses two pathways at ≥1 EC and adopts only 10.8% of the available nonzero sub-threshold ECs versus evw's 19.1%: confidence-independent pruning discards nonzero sub-threshold evidence that evw retains. The biomass floor (B2 vs. B3) barely changes detection, so the difference between these conditions is associated with the evidence-weighted penalty rather than the biomass floor.

The structural table isolates each component. B2 and B3 are near-equal in size (3,077

Table S5. Structural properties of each condition's MILP selected set, 108 genomes, DeepProZyme-vanilla. n_{selected}: reactions selected by the MILP (y>0.5). dead-ends/MI frac: per COBRA submodel (fraction failing elemental balance). Growing: non-zero FBA biomass under the defined medium.

Condition	n_{selected}	dead-ends ↓	MI frac ↓	Growing ↑
B2 (no biomass floor)	3,077	–	–	0/108
B3 (full METEOR, evw)	3,121	2.35	0.464	108/108
uniform (p=0)	2,215	1.74	0.492	108/108

Note. B2 vs. B3 (near-equal size, 0/108 vs. 108/108 growing) isolates the biomass floor. evw (B3) retains more reactions than uniform (3,121 vs. 2,215); uniform pruning does not use reaction confidence. B2's non-viable models are not MEMOTE-profiled (dead-ends/MI n/a).

vs. 3,121 selected reactions) yet zero of 108 B2 models grow, versus 108/108 for B3: the biomass floor requires a biomass-supporting route, making the selected submodels growth-feasible without substantially enlarging them. evw (B3) also retains more reactions than the uniform penalty (3,121 vs. 2,215). This comparison concerns reaction retention, not isozymes, because the formulation contains no protein-level variables.

Sequence-cost control: internal deletion-recovery. The deletion-recovery experiment (main text, Section 3.4) tests whether the sequence-derived cost changes selection when the feasible candidate set is held fixed. Associated EC-score columns are ablated, and the MILP is re-solved under the evidence-derived cost, a confidence-independent uniform cost, and shuffled costs. The evidence-derived cost selects 4–58× fewer off-reference reactions than the uniform or shuffled alternatives (Wilcoxon $p < 10^{-18}$, 108 genomes), relative to METEOR's unperturbed selected set. Because the reference is internal, this result measures stability rather than biological accuracy.

S3.1 Pathway coverage: the size-matched control

Set size confounds any pathway comparison between METEOR and a hard-threshold baseline, because METEOR's MILP-selected EC set is larger than the baseline's (≥ 0.5) set. Matching the sizes removes the confound directly. We therefore score METEOR's selected EC set of size K against the top-K SEED-representable ECs by maximum baseline score, on the same 151 pathways and the same 108 genomes (Table S6).

The result is negative. Against the fixed-threshold baseline METEOR gains 4.7–9.1 percentage points of mean pathway coverage. Against the size-matched baseline it loses 2.7–2.8 points, and falls below it in 107 of 108 genomes for CLEAN and in all 108 for DeepProZyme and EnzBERT. The apparent gain therefore reflects how many ECs METEOR returns, not better pathway recovery, and we make no pathway-inference claim in the main text.

For completeness we also record thresholded detection counts at three stringencies (Table S7). These favour METEOR, but they do not rescue a pathway claim: the least stringent detection measure is insensitive to how much of a pathway is present. One observation from that table does survive independently of the comparison baseline: 88–99% of the pathways ME-

Table S6. Size-matched pathway coverage on panel108 (151 SEED-representable pathways). thr = baseline at score ≥ 0.5; top-K = size-matched baseline, the K highest-scoring SEED-representable ECs with K = METEOR's selected EC count. Values are mean fraction of pathway ECs present. Δ columns are percentage points. lower in counts genomes where METEOR falls below the size-matched baseline.

Baseline	thr	top-K	METEOR	Δ vs thr	Δ vs top-K	lower in
CLEAN	0.168	0.242	0.215	+4.73	−2.69	107/108
DeepProZyme	0.214	0.332	0.305	+9.10	−2.70	108/108
EnzBERT	0.177	0.262	0.234	+5.74	−2.79	108/108

TEOR detects but the threshold baseline misses are associated with nonzero sub-threshold evidence ($0 < \text{score} < 0.5$) rather than zero-evidence reactions.

Table S7. Pathway detection counts (mean pathways/genome of 151) at three stringencies. A pathway is detected if the fraction of its ECs present is $\geq 1/n$ ($\geq$1 EC), $\geq$0.25, or $\geq$0.50. The $\geq$5-EC cutoff behind the 151 is what keeps these three apart: a pathway with four or fewer SEED-representable ECs reaches $\geq$0.25 on a single EC, so that column would repeat the $\geq$1 EC one. Relaxing the cutoff to $\geq$3 or $\geq$1 admits 162 and 166 pathways and changes no conclusion. The $\geq$1 EC gain runs +5.9 to +9.7 across all three settings, the $\geq$50% ratio 1.79–2.17×, and the nonzero sub-threshold share of METEOR-only detections stays at 88–99%. M-only weak/gap: of pathways METEOR detects ($\geq$1 EC) that the baseline misses, the split between nonzero sub-threshold driving ECs ($0 < \text{score} < 0.5$) and zero-evidence ECs.

Baseline	$\geq$1 EC $B{\to}M$	$\geq$25% $B{\to}M$	$\geq$50% $B{\to}M$	M-only (weak/gap)
CLEAN	107.6→114.1	42.0→55.8	10.2→20.0	8.3/0.1
DeepProZyme	123.9→132.5	56.3→80.2	16.6→36.0	9.1/1.3
EnzBERT	115.4→121.5	43.7→61.3	10.4→22.1	7.6/0.1

The additional selection is limited rather than indiscriminate: of the ECs with nonzero sub-threshold evidence ($0 < \text{score} < 0.5$) available per genome, METEOR selects only 308/4146 (7.4%, CLEAN), 328/1715 (19.1%, DeepProZyme), and 334/3711 (9.0%, EnzBERT); and of the pathways that carry recoverable sub-threshold evidence (baseline-missed but with $\geq$1 such EC), it recovers only 19.3% (8.4/43.4, CLEAN), 39.3% (9.5/24.1, DeepProZyme), and 21.7% (7.7/35.6, EnzBERT). METEOR therefore does not admit every nonzero-scored EC; the evw cost prunes the large majority of that evidence and the network selects a small, biomass-associated subset (`eval/kegg_select.py`, `kegg_pathway_108_selectivity.json`).

S4. Alternative repair

To test whether simpler repair methods can recover the structural quality that METEOR achieves, we compared two alternatives on the same 108-GCF panel under the same defined growth medium used by METEOR (29 SEED compounds plus 7 essential inorganics; 36 importable exchanges total).

Greedy topology-only repair. Starting from the DeepProZyme-vanilla noisy-OR threshold network (mean 6,704 selected reactions at the noisy-OR threshold $w_j \geq 0.5$; the corresponding COBRA submodel, which adds compartment and exchange reactions, is an 11,637-reaction network), dead-end fraction 0.789, we iteratively add the lowest-log-odds-cost unselected reaction touching a dead-end metabolite, capping at 20,000 iterations. The greedy algorithm adds a mean of 16,445 reactions yet plateaus at dead-end fraction 0.235, because each new reaction introduces fresh dead-ends at its stoichiometric boundary. Zero of 108 genomes support biomass growth (all hit the iteration cap; biomass flux $= 0$), and the procedure takes a median of 64 min per genome.

Comparison against a general-purpose gap-filler. An earlier implementation omitted the medium from the thresholded draft, which has no uptake reactions of its own. We discarded its invalid zero-growth output and carried the shared medium into both arms of the corrected comparison below.

The corrected comparison fixes everything except the algorithm. Both arms start from the same DeepProZyme threshold draft (median 2,892 reactions), solve under the same medium, and draw from one shared candidate set R (median 20,156 of the universal's 47,880). We build R without consulting METEOR's confidences, using EC-annotated reactions, the biomass-feasible skeleton, and the medium exchanges, less those structurally excluded. The candidate set therefore does not use METEOR's reaction-confidence costs.

The formulations differ, and that is the point. METEOR repairs with a linear program minimising flux through non-core reactions; COBRApy's[1] `GapFiller` solves a mixed-integer program minimising the number of reactions added, one binary variable per candidate. Across 108 genomes the LP returned a growing solution every time, in a median 9.6 s, adding a median of 82 reactions. The MILP returned no solution within a 3,120 s budget in any genome, about 330× the LP's median, and was faster in no genome.

Two readings this does not support. The MILP is not shown to be unsolvable here: it was cut off and never proved infeasible, so its runtime and the runtime ratio are both lower bounds. Neither is the LP shown to be the better algorithm, since it solves an easier problem; on a 95-reaction textbook model, where both finish in 0.03 s, the two return the same minimal set of four reactions. The finding concerns scale alone. Within this time budget, minimum-count gap-filling does not reach a solution on a candidate set this size, which is why METEOR repairs with a flux-parsimony LP. The per-genome `results/gapfill_paired/` records behind this comparison are not deposited, so it cannot be re-read from the deposited records. Both scripts are in the release: `eval/cgf2.py` produces the records and `eval/agg_gapfill.py` aggregates them.

Holding the predictor fixed gives a sharper comparison still: Section S7.4 gap-fills the same DeepProZyme threshold draft with the parsimony repair METEOR uses internally, and those models grow. That is the arm against which evidence-weighted selection shows no accuracy advantage (EC-level recall 0.784 versus 0.767). The deletion-recovery experiment includes a separate two-stage control constructed on its fixed candidate set.

S5. Parameter sensitivity

We assess sensitivity to the three hyperparameters that most directly shape METEOR's output: the score-truncation depth k (input), the confidence-update strength α (per-protein), and the evidence-weighted-parsimony penalty (p, μ) (network). We fixed the remaining hyperparameters ($\gamma_{\max} = 2.5$, $\lambda = 10^{-4}$, $\epsilon = 10^{-6}$, $\delta = 0$, aggregation = noisy-OR) at their production values and froze them before holdout evaluation; $\gamma_{\min}$ is treated separately below. A full Cartesian sweep of every parameter was not performed; we report the three available sweeps and the frozen defaults for the remaining parameters.

Score-truncation depth $\boldsymbol{k}$. Each protein keeps its k highest-scoring ECs before the noisy-OR (main text, Section 2.2). We swept $k \in \{1, 3, 5, 10\}$ and no truncation across 53 genomes with the log-odds cost alone, without the evidence-weighted parsimony term $\mu(1-w_j)^p$. The launcher is deposited as `eval/legacy/memote_ksens.sbatch`; the per-genome records are not, so the table documents an earlier sensitivity analysis rather than a reproducible analysis of the final formulation.

Table S8. Sensitivity to the truncation depth k. Protein-level metrics are computed on the legacy 67-protein holdout; selected reactions is the mean over 53 genomes. This sweep was run with the log-odds cost alone, so the reaction counts are larger than the evidence-weighted counts reported in the main text.

k	top-1	top-3	$F_{\max}$	selected reactions
1	18/67	23	0.317	4,819
3	18/67	23	0.317	5,932
5 (default)	18/67	23	0.317	6,596
10	18/67	23	0.317	7,509
none	18/67	23	0.318	9,607

The protein-level output does not move at all (Table S8): top-1 and top-3 are identical at every k, and $F_{\max}$ shifts by 0.001 only when truncation is removed entirely. The selected reaction count, in contrast, rises monotonically and roughly doubles between k=1 and no truncation. We therefore describe k as a parsimony control rather than an accuracy control, and we do not claim the method is insensitive to it: k decides how large the reconstructed model is in this preceding formulation. The default k=5 was fixed before holdout evaluation; this sweep does not identify an accuracy optimum.

Confidence-update strength $\boldsymbol{\alpha}$. We swept $\alpha \in \{0.2, 0.4, 0.6\}$ (equivalently $\beta = \alpha/0.4 \in \{0.5, 1.0, 1.5\}$) on the 66 net-only-evaluable holdout proteins (DeepProZyme baseline, production p=2, μ=3 MILP). α rescales the posterior update only; the MILP solution is identical across α, so both the baseline and METEOR scores are mapped through the same EC-vocabulary normalisation (net-only) as elsewhere.

Table S9. Sensitivity of net-only per-protein metrics to the confidence-update strength α (equivalently $\beta = \alpha/0.4$) on n=66 holdout proteins (hard-biomass MILP, DeepProZyme baseline, production p=2, μ=3). α=0.4 is the production default.

α (β)	n	top-1 ↑	top-3 ↑	$F_{\max}$ ↑
0.2 (0.5)	66	17	22	0.308
0.4 (1.0)	66	17	22	0.307
0.6 (1.5)	66	17	22	0.302

Top-1 and top-3 are exactly invariant across the range (Table S9; the per-EC update never crosses a first-to-second-EC gap); $F_{\max}$ varies by ≤ 0.006 and is highest at the most conservative α=0.2. The production default α=0.4 was frozen before holdout evaluation and is not the $F_{\max}$-optimal value, consistent with the statement that it was not tuned to the holdout. Calibration metrics under the same update are reported separately in Table S2.

Parsimony penalty $(\boldsymbol{p}, \boldsymbol{\mu})$. The exponent p and weight μ of the penalty $\mu(1-w_j)^p$ govern how aggressively low-confidence reactions are pruned, so their natural readout is network-level. We evaluate the full $p \in \{0, 1, 2, 3\} \times \mu \in \{1, 3, 5, 8\}$ grid against reaction-level references from the six curated GEMs, the only cohort with curated reaction sets, reporting MetaNetX-bridged[2] precision, recall, and Jaccard of the repaired selected set against the curated model, plus mean network size.

Reaction overlap with the curated models is nearly flat across the entire grid (Table S10; Jaccard 0.12±0.01): larger μ, or the confidence-independent p=0, prunes more aggressively and trades recall for precision at roughly constant Jaccard. The production default (p=2, μ=3) is not the Jaccard- or precision-optimal cell; uniform p=0, μ=5 attains the highest Jaccard. The default was therefore not chosen to maximise either displayed metric, and evidence weighting does not improve static curated-model overlap on this grid. Its measured effect appears instead in the internal deletion-recovery comparison (main text, Section 3.4). Retaining a redundant route that the evidence supports adds a reaction the curated reference may not list, so higher overlap is not what the penalty targets.

Biomass floor $\boldsymbol{\gamma_{\min}}$. The remaining MILP parameter of interest is $\gamma_{\min}$, the hard lower bound on biomass flux. The ablation in Section S3 provides a direct comparison: $\gamma_{\min}$=0 (condition B2) vs. $\gamma_{\min}$=0.1 (condition B3, the production default). At the protein level, both conditions yield 17/66 top-1 and $F_{\max}$ = 0.307. Their pathway-detection counts are also close (132.2 vs. 132.5 pathways with at least one EC; 35.3 vs. 36.0 at least half complete). The structural result

Table S10. Sensitivity of curated-GEM reaction recovery to the parsimony penalty (p, μ), mean over the six curated-GEM model organisms (DeepProZyme baseline; p=0 is the confidence-independent uniform penalty). (p=2, μ=3) is the production default.

p	μ	Prec.	Recall	Jaccard	n_{selected}
0	1	0.173	0.286	0.120	3188
0	3	0.200	0.266	0.128	2580
0	5	0.222	0.237	0.129	2116
0	8	0.243	0.185	0.117	1532
1	1	0.165	0.290	0.116	3378
1	3	0.171	0.287	0.119	3222
1	5	0.178	0.284	0.122	3084
1	8	0.181	0.281	0.123	2994
2	1	0.163	0.292	0.116	3439
2	3	0.166	0.290	0.117	3360
2	5	0.168	0.289	0.118	3305
2	8	0.171	0.288	0.119	3249
3	1	0.162	0.293	0.115	3472
3	3	0.163	0.292	0.116	3418
3	5	0.165	0.291	0.117	3387
3	8	0.166	0.290	0.117	3354

Note. n=6 organisms; the production default is p=2, μ=3. The raw sweep directory is not included in the release.

differs: B2 produces 0/108 growth-feasible submodels, B3 produces 108/108. This ablation shows that the production biomass floor supplies feasibility; it does not test sensitivity among positive values of $\gamma_{\min}$. We fixed $\gamma_{\min}$=0.1 during earlier development and froze it before holdout evaluation. Because the SEED biomass reaction is normalised, $\gamma_{\min}$=0.1 is a minimal viability floor, not a quantitative growth-rate prediction.

S6. Additional baselines

The main text's pathway-detection analysis (Section 3.3) and the MEMOTE structural quality profiles (Supplementary Section S7) cover the three baselines with full 108-GCF-panel predictions (CLEAN, DeepProZyme, EnzBERT). Three additional baselines (GraphEC,[3] a structure-aware graph neural network; MAPred;[4] and TopEC[5]) are evaluated on the smaller 22-genome Price-149 panel (n=149 proteins with experimentally verified EC annotations across 22 organisms). Price et al.[6] determined those annotations experimentally; ProteInfer[7] first assembled them as a benchmark, on the grounds that automated methods had labelled these sequences incorrectly or inconsistently in databases such as KEGG. Table S11 reports per-protein fidelity; Table S13 reports pathway completeness. Organism-scale predictions for these three baselines exist only on this smaller panel.

Under the net-only comparison, METEOR leaves top-1 nearly unchanged (CLEAN +1,

Table S11. Per-protein fidelity on the Price-149 benchmark (n=149, 22 genomes, hard-biomass MILP), net-only (baseline and METEOR mapped through the identical EC-vocabulary normalisation). B = baseline; $+M$ = METEOR-refined. All six baseline architectures, including the structure-aware GraphEC, are evaluated under the same MILP configuration.

	top-1 ↑		$F_{\max}$	
Baseline	B	$+M$	B	$+M$
CLEAN	60	61	0.559	0.539
DeepProZyme	49	49	0.558	0.529
GraphEC	18	18	0.264	0.233
MAPred	50	52	0.550	0.537
TopEC	5	5	0.034	0.034
EnzBERT	53	53	0.477	0.463

MAPred +2, the rest unchanged; no top-1 regression) but does not improve $F_{\max}$: $\Delta F_{\max}$ ranges from -0.031 (GraphEC) to 0 (TopEC). The earlier apparent gains (e.g. GraphEC +0.030) were an EC-vocabulary artifact of a non-net-only comparison. The two largest $F_{\max}$ declines occur for GraphEC and DeepProZyme, for which thousands of ECs per protein are muted (Table S12). TopEC, which scores essentially one EC per protein, is unchanged; CLEAN is an exception to a simple support-size explanation because only 37 of its many nonzero ECs are muted. Top-1 changes only for CLEAN and MAPred. We therefore do not claim METEOR improves out-of-distribution protein-level fidelity; on this benchmark the network-selection step leaves top-1 nearly unchanged while imposing a small, predictor-dependent $F_{\max}$ cost.

Table S12. Output-distribution profile of the six predictors on Price-149 and the resulting net-only $\Delta F_{\max}$. Support = mean number of ECs scored > 0 per protein; conf. = fraction of proteins whose top EC scores ≥ 0.5; muted = mean ECs demoted by METEOR per protein. These counts contextualise the observed change in $F_{\max}$.

Predictor	Support	conf.	muted/prot	$\Delta F_{\max}$
GraphEC	5,071	0.34	5,068	−0.031
CLEAN	5,236	0.59	37	−0.020
DeepProZyme	2,715	0.73	2,674	−0.029
EnzBERT	100	0.58	97	−0.014
MAPred	27	0.93	22	−0.013
TopEC	1	0.74	0.07	0.000

We repeat the pathway analysis of Section 3.3 for the three additional predictors on the

22 Price-149 genomes, using the same P=151 SEED-representable pathways and reporting the same detection-and-matched-completeness metrics rather than raw average completeness (which is confounded by reaction-set size; Section 3.3, Supplementary Section S3).

Table S13. Price-149 pathway detection for GraphEC, MAPred, and TopEC (22 genomes, 151 pathways). Detection: mean pathways/genome with ≥1 EC / ≥50% of ECs present ($B \to M$). M-only (weak/gap): pathways METEOR detects (≥1 EC) that the baseline misses, split into nonzero sub-threshold ($0 < \text{score} < 0.5$) vs zero-evidence driving ECs.

Baseline	det. ≥1 EC $B \to M$	det. ≥50% $B \to M$	M-only (weak/gap)
GraphEC	75.1→102.6	4.3→12.5	29.4 (29.0/0.4)
MAPred	134.8→134.4	46.8→57.7	3.0 (2.7/0.3)
TopEC	107.6→112.8	1.7→8.4	6.4 (1.2/5.1)

The descriptive pattern matches the main panel (Section 3.3): pathway detection at ≥50% completeness increases for all three, and at ≥1 EC for GraphEC and TopEC, most strongly for the diffuse structure-aware GraphEC (+27.5 pathways at ≥1 EC, ∼3× at ≥50%); MAPred edges down at ≥1 EC (134.8 → 134.4) while its ≥50% count rises (46.8 → 57.7), and 98.6% of GraphEC's METEOR-only detections are driven by nonzero sub-threshold evidence (29.0/29.4), and 90% for MAPred. TopEC is an exception: it commits to essentially one EC per protein (Table S12), so it carries almost no sub-threshold signal to select; 80% of its METEOR-only detections (5.1/6.4) come from zero-evidence gap-fill reactions. For predictors with graded scores, most additional detections are associated with nonzero sub-threshold evidence; for TopEC, they predominantly reflect gap-filling. As in the main analysis, these detection counts do not establish pathway recovery.

S7. Structural quality, external reconstruction context, and computational cost

S7.1 MEMOTE structural quality and comparison with CarveMe

We compare METEOR and CarveMe[8] on MEMOTE structural quality under their respective reconstruction settings. The two tools differ in every design dimension: input (bounded EC scores vs. genome), template (SEED universal vs. curated BiGG), namespace, and objective. This is context rather than a controlled comparison. METEOR's mean MEMOTE total score is higher in these records for all three baselines in the vanilla configurations (0.765–0.788 vs. 0.735–0.739 for CarveMe; 0.765–0.796 across all twelve), as are its dead-end, mass-balance, and charge-balance sub-scores. METEOR has a higher unbounded-flux fraction (mean 0.75–0.94 in the vanilla configurations, 0.73–0.94 across all twelve, vs. 0.545; lower is better), a metric that favours CarveMe's smaller, manually curated BiGG template.

Every reported METEOR selected set attains the biomass floor after post-solve verification and repair (108/108; main text, Section 3.1). A downstream FBA on the MEMOTE submodel uses an uptake medium inferred post hoc, which differs from the MILP medium (Section S1), and grows in 105–108/108 genomes in the three vanilla configurations (108/108 for DeepProZyme and EnzBERT; 105/108 for CLEAN). Across all twelve configurations the range is 83–108/108, the three lowest being CLEAN-Filt70 (83/108), CLEAN-Filt30 (94/108) and CLEAN-Filt50 (103/108). The difference concerns the downstream medium and submodel conversion; it does not overturn the fixed-medium post-solve check.

Neither tool's structural scores should be interpreted as protein-level accuracy: CarveMe over-predicts amino-acid auxotrophies,[9] and automated pipelines including CLEAN reach only 50–69% accuracy on experimentally verified enzymes.[10] MEMOTE evaluates network-level structural quality, not whether each predicted EC is correct. Namespace-neutral EC-level recall against curated GEMs, the comparison we do rely on in the main text, is reported in Sections S7.4 and S7.5.

S7.2 MEMOTE sub-score breakdown

Table S14 reports the per-section MEMOTE scores across all 12 baseline×cutoff configurations on the 108-GCF panel. Consistency (weight 3) is the only sub-score that varies meaningfully across configurations (84.5–88.1%), driven by the number and composition of reactions selected by the MILP. The four annotation sub-scores are nearly uniform because they derive from the shared SEED universal template. CLEAN-Vanilla shows an anomalous gene-annotation score (27.8% vs. 40.0% elsewhere) due to 33 genomes with zero gene annotations in the COBRA submodel; this artefact does not affect the consistency or total-score ranking. The two rightmost columns report, per configuration, the fraction of the 108 genomes with non-zero FBA-optimal biomass flux under the post-hoc inferred downstream medium (Growing) and the mean MEMOTE `test_find_reactions_unbounded_flux_default_condition` metric (Unbounded, fraction of reactions carrying unbounded flux in the default condition; lower is better); both are extracted from the same per-genome MEMOTE snapshot reports underlying the sub-scores at left. Growing ranges 76.9–100.0% ($\geq$ 97% in 9 of 12 configurations); Unbounded ranges 0.73–0.94, versus CarveMe's 0.545 on the same 108-genome panel (Section S7.1).

S7.3 Computational cost

We measured aggregation and MILP construction while subsampling the SEED universal from 10^3 to 47,880 reactions. Aggregation increases from 0.02 to 0.29 s and construction from 0.06 to 5.9 s. Solve time dominates because the optimiser must choose a biomass-feasible candidate subnetwork from the universal.

On the 108-genome panel at the configuration this paper reports (DeepProZyme-vanilla, evidence-weighted cost, δ=0), the MILP solve together with model construction takes a median of 137 s per genome and a mean of 211 s, ranging from 72 to 738 s, on four CPU cores of an Intel Xeon Gold 6242. These are the wall times recorded by the solver-status run in Section S1.1; they cover one configuration, not the twelve.

For scale against another tool we have only the earlier measurement: across the 109-genome panel and all twelve baseline×cutoff configurations, METEOR's median per-genome

Table S14. MEMOTE quality sub-scores for reconstructed models across the 108-GCF panel (n=108 genomes per configuration). Consist.–Total values are mean $\pm$ std (%); MEMOTE section weights shown in parentheses; Total is the weighted aggregate. Growing: fraction of genomes with non-zero FBA-optimal biomass flux under the post-hoc inferred downstream medium. Unbounded: mean fraction of reactions carrying unbounded flux (MEMOTE consistency test, lower is better). Bold marks the highest value in the Consistency and Total columns.

Baseline	Cutoff	Consist. (w=3)	Ann. Met. (w=1)	Ann. Rxn. (w=1)	Ann. Gene (w=1)	Ann. SBO (w=2)	Total	Growing (%)	Unbounded
CLEAN	Vanilla	87.5 ± 1.5	68.5 ± 0.1	65.1 ± 0.3	27.8 ± 18.4	76.2 ± 8.4	76.5 ± 4.5	97.2	0.75
	Filt-30	88.1 ± 2.0	68.5 ± 0.1	65.1 ± 0.2	39.6 ± 3.8	81.6 ± 1.7	79.6 ± 1.1	87.0	0.73
	Filt-50	88.0 ± 1.7	68.5 ± 0.2	65.1 ± 0.7	40.0 ± 0.0	81.7 ± 0.9	79.6 ± 0.8	95.4	0.73
	Filt-70	86.8 ± 1.8	68.5 ± 0.1	65.2 ± 0.2	40.0 ± 0.0	81.8 ± 0.0	79.2 ± 0.7	76.9	0.81
DPZ	Vanilla	84.5 ± 0.1	68.4 ± 0.0	65.7 ± 0.0	40.0 ± 0.0	81.8 ± 0.0	78.3 ± 0.0	100.0	0.94
	Filt-30	85.3 ± 1.1	68.4 ± 0.1	65.4 ± 0.2	40.0 ± 0.0	81.8 ± 0.0	78.6 ± 0.4	99.1	0.88
	Filt-50	85.1 ± 1.2	68.4 ± 0.1	65.4 ± 0.3	40.0 ± 0.0	81.8 ± 0.0	78.6 ± 0.4	99.1	0.88
	Filt-70	85.3 ± 1.6	68.4 ± 0.1	65.4 ± 0.2	40.0 ± 0.0	81.8 ± 0.0	78.6 ± 0.6	100.0	0.87
EnzBERT	Vanilla	85.7 ± 1.9	68.4 ± 0.1	65.3 ± 0.2	40.0 ± 0.0	81.8 ± 0.0	78.8 ± 0.7	100.0	0.84
	Filt-30	85.5 ± 2.0	68.4 ± 0.1	65.4 ± 0.2	40.0 ± 0.0	81.8 ± 0.0	78.7 ± 0.8	99.1	0.84
	Filt-50	85.4 ± 1.9	68.4 ± 0.1	65.4 ± 0.2	40.0 ± 0.0	81.8 ± 0.0	78.7 ± 0.7	100.0	0.86
	Filt-70	85.1 ± 1.8	68.4 ± 0.1	65.4 ± 0.2	40.0 ± 0.0	81.8 ± 0.0	78.6 ± 0.7	98.1	0.87

build (solve, submodel construction, and MEMOTE evaluation together) was 6.7 minutes on four cores against CarveMe's 20.3 minutes on two, so roughly three times faster in wall-clock and about 1.5× cheaper in core-time. That comparison predates the move to the 108-genome panel and we have not repeated it; we report it as an order-of-magnitude statement rather than a matched benchmark.

The reported METEOR run uses the open-source CBC solver. Per-protein EC inference is a separate GPU pre-step and is not included in these timings. METEOR parallelises across genomes by SLURM array.

S7.4 The same-predictor threshold-and-gap-fill baseline

Sections 3.5 and S7.5 compare METEOR against a baseline built from the same predictor. The comparison holds the score source and biochemical database fixed, but the methods use different selection objectives and candidate masks. We give the construction here because several results depend on it.

Draft. For each EC we take the maximum score over the proteome (DeepProZyme, vanilla). A reaction enters the draft if any EC annotated to it reaches τ=0.5. This is the hard-threshold reading of the same scores METEOR consumes.

Gap-fill. The draft alone does not grow (main text Table 1: 0/108 for every predictor), so we complete it with the parsimony repair METEOR uses internally, `repair.grow_support`: a linear program over the reactions that can carry flux under the Gram-specific bounds, which holds $\mathbf{Sv} = \mathbf{0}$ and $v_{\text{biomass}} \geq 0.1$ while minimising flux through reactions outside the draft. The draft is therefore reused at no cost and only bridging reactions are added. Should that prove infeasible, we retry over the unrestricted universal.

Shared components and remaining differences. Both arms use the SEED universal, Gram-specific tight bounds, defined medium, biomass floor $\gamma_{\min} = 0.1$, and CBC. The baseline's repair LP can draw from all non-excluded reactions and minimises flux outside the thresholded draft.

METEOR restricts candidates using $w_j \geq 0.01$, the medium, and the shared biomass skeleton, then charges evidence-derived cost once per selected binary reaction. The comparison therefore evaluates the complete two-stage baseline against the complete METEOR selection procedure; it does not isolate a single algorithmic component.

Across the 25 genomes on which the two arms were compared, the gap-fill adds a mean of 85 reactions (range 54–253) and both arms grow in all 25. The EC-level recall figures quoted in Section 3.5 come from the six of those genomes with a published curated model. Code `eval/baseline_thresh_gapfill.py`; results `results/toolcompare/ablation_thresh_vs_evw{,_ec}.json`.

This baseline is distinct from the COBRApy `GapFiller` timing comparison in Section S4: it uses METEOR's repair LP, and all resulting models grow.

S7.5 Curated sub-threshold EC recovery

This section supports the main-text result of Section 3.2 with per-organism counts, denominators and interval estimates. Its reference is a published curated model, so unlike the pathway analysis it does not depend on how a pathway is defined or on how many ECs a method returns.

For each genome let W be the set of ECs that the curated model contains and that the predictor scores in $(0, 0.5)$. These ECs are present in the curated model but invisible to a hard threshold. Averaged over the six genomes $|W| = 102$ ECs. We ask what fraction of W each method ends up including.

Table S15. Recovery of curated-reference sub-threshold ECs, DeepProZyme scores, six curated GEMs. Recovery is the fraction of W the method includes; precision is against the full curated EC set. Per-organism and pooled intervals are Wilson 95% intervals on the recovered proportion; the mean interval is a bootstrap over organisms. The baseline is the same-predictor threshold-and-gap-fill arm reported in main text Section 3.2.

Organism	$\|W\|$	baseline	METEOR	METEOR 95% CI
E. coli	93	0.495	0.656	$[0.555, 0.745]$
Salmonella	84	0.452	0.560	$[0.453, 0.661]$
K. pneumoniae	90	0.478	0.633	$[0.530, 0.726]$
P. putida	119	0.479	0.613	$[0.524, 0.696]$
S. aureus	123	0.431	0.520	$[0.433, 0.607]$
B. subtilis	103	0.544	0.641	$[0.545, 0.727]$
Mean recovery	102	0.480	0.604	$[0.562, 0.640]^{\dagger}$
Pooled recovery	612	0.479	0.601	$[0.562, 0.639]$
Mean precision	–	0.455	0.422	–

†Bootstrap over the six organisms, 10^4 resamples, seed 0; the pooled row uses a Wilson interval on 368/612. Baseline pooled recovery is 293/612.

METEOR recovers 0.604 of W against the baseline's 0.480 (Table S15), higher in all six genomes (p=0.031, the smallest value six paired samples admit). Precision falls by 0.032, also in all six. The threshold discards every sub-threshold EC from the draft; METEOR's complete selection procedure includes a further twelve percentage points of them and pays for it in precision. Because the objectives and candidate masks differ, this comparison does not attribute the difference to the cost term alone.

Recovered functions. The ECs METEOR recovers and the baseline misses include specific metabolic capabilities as well as questionable matches, illustrating both the signal and the metric's limits. We checked every example below against the ENZYME[11] flat files (`https://enzyme.expasy.org/EC/`) and KEGG (`https://rest.kegg.jp/get/ec:`); all are current EC numbers, none transferred or withdrawn. Counts are the organisms in which the EC is recovered by METEOR only, out of those in which it is a curated sub-threshold EC.

- EC 1.1.1.133, dTDP-4-dehydrorhamnose reductase (2/2): supplies dTDP-L-rhamnose for O-antigen assembly. The official equation is written in the oxidative direction although the enzyme operates as a reductase within the dTDP-L-rhamnose synthase complex.
- EC 1.17.1.4, xanthine dehydrogenase (1/1): purine catabolism; distinct from EC 1.17.3.2 (xanthine oxidase).
- EC 1.2.1.21, glycolaldehyde dehydrogenase (2/6): its sole cross-referenced protein in ENZYME is P25553 (ALDA_ECOLI), the E. coli enzyme.
- EC 1.3.8.1, short-chain acyl-CoA dehydrogenase (3/4) and EC 1.3.3.4, protoporphyrinogen oxidase (2/3): fatty-acid β-oxidation and haem biosynthesis.
- EC 1.3.8.9 (very-long-chain acyl-CoA dehydrogenase) and EC 1.3.3.6 (acyl-CoA oxidase) are recovered 3/3 each, but both are more typical of eukaryotic β-oxidation and we do not rest anything on them.

One recovery is worth flagging against ourselves: EC 2.1.1.37, DNA (cytosine-5-)-methyltransferase, recovered in B. subtilis, is a DNA-modification rather than an intermediary-metabolism enzyme, reaching KEGG metabolism through its S-adenosyl-L-methionine cofactor. It is in the curated model's EC annotation and so counts as a recovery under this metric, which illustrates the metric's limit: it scores repertoire membership, not contribution to the network.

Two limits on this result. Six genomes cannot establish more than a direction, and it was measured on DeepProZyme alone, so we do not know whether the margin holds for predictors whose score distributions differ (Section 3.1 shows those distributions vary widely). Code `eval/weakreal.py`; results `results/toolcompare/weakreal.json`.

S8. Full-proteome protein-level comparison

To examine protein-level fidelity beyond the 66-protein holdout set, we evaluated METEOR on all proteins carrying four-digit EC annotations in the NCBI RefSeq records (GenBank flat file format) of the 95 genome-collection assemblies that carry such annotations (94 for the CLEAN-filt30 configuration). For each genome, we downloaded the GBFF file via the NCBI Datasets API, extracted `/EC_number` qualifiers from CDS features, and matched proteins by their RefSeq accessions (NP_/WP_/YP_) to the baseline and METEOR prediction DataFrames.

Scope and limitation. Unlike the primary holdout (which post-dates all baseline training data), these NCBI-annotated proteins are not held out; many appear in baseline training sets. Absolute accuracy is therefore not comparable to the holdout; the baseline-vs-METEOR comparison remains valid because both methods receive identical input scores.

Under the fair net-only comparison, the baseline is mapped through the same EC vocabulary as `meteor_df`, so that the vocabulary normalisation is not mis-credited to METEOR. The score feedback does not improve top-1 accuracy on the full proteome. For DeepProZyme-vanilla (n=69,666), top-1 is 90.2%→90.1% (103 corrections vs. 188 regressions) and $F_{\max}$ is unchanged; the only positive effect is a small decision-threshold recovery (micro-F_1@0.5 +0.002). This is consistent with the holdout (F_1@0.5 +0.008; top-1 unchanged and $F_{\max}$ nearly unchanged) and shows little change in per-protein performance.

S8.1 Net-only correction rate by training-set sequence identity

Stratifying the net-only top-1 outcome by each protein's maximum identity to the DeepProZyme training set (DIAMOND[12]) shows the per-protein effect is positive by three proteins in the lowest-identity bin and negative in every other bin (Table S16). This descriptive difference does not establish that the metabolic constraint helps low-identity proteins. The inflated correction counts of earlier non-net-only comparisons were an artifact of an unnormalized baseline vocabulary.

Table S16. Net-only top-1 corrections/regressions by maximum training-set sequence identity (DeepProZyme, n=69,666). Net = corrections − regressions.

Identity	n	top-1 B	top-1 $+M$	Corr	Regr
<30%	2,911	73.8%	73.9%	21	18
30–50%	17,013	87.4%	87.3%	38	64
50–70%	15,838	93.2%	93.0%	8	28
70–90%	11,234	95.1%	95.0%	1	11
>90%	19,801	95.5%	95.4%	6	19
no hit	2,869	52.0%	51.4%	29	48

S9. Internal deletion-recovery stability

On the full 108-genome panel (63 Gram-negative and 45 Gram-positive; no genome subselection), we test how the evidence-derived cost affects selection beyond biomass feasibility. Each genome's unperturbed METEOR selected set is the internal reference. Using `RandomState(0)`, we sample D=60 selected, evidence-bearing, non-skeleton reactions, collect every EC associated with them, and set those EC-score columns to zero for all proteins. Because an EC can annotate several reactions, the perturbation can remove evidence from reactions beyond the sampled 60.

We recompute costs and solve over an identical candidate set under four regimes: the organism-specific evidence-derived cost, a two-stage threshold-then-weighted-gap-fill procedure, a confidence-independent uniform cost, and shuffled evidence-derived costs. The metric is the number of off-reference reactions selected, meaning reactions absent from the unperturbed METEOR set.

Table S17. Internal deletion-recovery stability on the full 108-genome panel (mean off-reference reactions per genome), overall and split by Gram type.

regime	all (n=108)	Gram− (n=63)	Gram+ (n=45)
evidence-derived	34.6	31.2	39.4
two-stage	78.2	70.1	89.6
uniform	142.9	133.8	155.6
shuffled	2,006.6	1,958.7	2,073.6

All four regimes satisfy the biomass constraint but select different sets (Table S17). Relative to the evidence-derived cost, the mean off-reference count is 2.3× higher for two-stage, 4.1× higher for uniform, and 58.0× higher for shuffled costs. Recovery precision, the fraction of newly selected reactions that belong to the sampled set D, is low throughout: 0.037 for the evidence-derived cost, 0.022 for two-stage, 0.008 for uniform, and 0.004 for shuffled costs. The two-stage arm also differs in objective shape: it shifts costs to be positive and charges them per unit flux, whereas METEOR charges the unshifted cost once per selected binary reaction. It therefore does not isolate architecture alone.

This experiment measures stability relative to METEOR's own output, not biological accuracy. It uses one fixed sample per genome, and the deposited JSON files do not retain the identities of the sampled reactions. Code: `eval/recovery_ablation.py`; per-genome results: `results/recovery_4arm/*.json`.

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